# Murray S. Bennett

The Ohio State University
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**Research Summary and Interests**

I am a cognitive psychologist using mathematical models, behavioural experiments, and machine learning to study perception, workload capacity, and decision making in people and teams. My work includes uncertainty estimation in AI systems and computational methods for testing cognitive theories. I aim to connect these theoretical and methodological advances with practical tools, including research on human–AI collaboration.

## EDUCATION

**Doctor of Philosophy in Psychology – Science** *2023*
University of Newcastle, Australia
Collective Workload: Human Performance in Competitive and Collaborative Tasks

**Bachelor of Psychology (Honours), Class I** *2018*
University of Newcastle, Australia
Honours Thesis: The Mental Representation of Symbolic Numerals

## RESEARCH

**Postdoctoral Scholar** *2024-*
The Ohio State University, Department of Psychology

- Develop computational methods for communicating neural network metacognition to human users, with the goal of improving human–AI collaboration and decision-making.
- Apply advanced quantitative cognitive models (including Systems Factorial Technology and General Recognition Theory) to test processing independence and interactions in social and cognitive decision-making tasks.
- Research conducted within NSF-funded projects (SES-2237119; SES-2523723).

**Postdoctoral Researcher** *2022-24*
UT San Antonio, Psychology Department

- Quantified human perceptual judgements and combined them with computer-vision estimates to improve classification in an offline melanoma image dataset.
- Used General Recognition Theory to examine perceptual interactions and feature integration in medical image decision-making.

**Data Analyst** *2021-22*
Hunter New England Health Local Health District

- Conducted weekly statistical analyses of nationwide FluTracking and COVID-19 survey data for Australian federal government reporting.
- Collaborated with international partners to design, implement, and maintain large-scale survey infrastructure for data collection in New Zealand and Argentina.
- Role held concurrently with PhD candidature.

**Research Assistant** *2021*
University of Newcastle, School of Biomedical Sciences and Pharmacy

- Designed, conducted, and analysed experiments examining physiological correlates of cognitive workload using the Detection Response Task.

**Research Assistant** *2019-22*
University of Newcastle, School of Psychology

- Provided technical and methodological support for research on cognitive processing, decision-making, and perceptual judgement.
- Designed and implemented online versions of foundational cognitive psychology experiments for research and undergraduate instruction.

**Research Internship** *2018–19*
University of Newcastle, Centre for Brain and Mental Health Research

- Analysed cognitive workload and flight-simulator performance data from helicopter pilots evaluating novel 3D helmet display systems.
- Examined relationships between task demands, performance, and subjective workload in applied human-factors contexts.

## TEACHING

**Guest Lecturer**
Artificial Intelligence in Psychology *OSU 2026*

- Lectured on k-means clustering and supervised student-led presentations.

**Course Instructor**
Introduction to Psychology *UT San Antonio 2024*

- Introduction to the study of and scientific approaches to attention, sensation, perception, emotion, learning, memory, problem solving, personality, development, atypical behaviour, and social behaviour.

**Honours Thesis Grader**
University of Newcastle, School of Psychology *UoN 2021*

- Assessed and critiqued the methodology, data analysis, and scientific writing of honours-level cognitive psychology research projects.

**Lab Demonstrator**
Advanced Research Methods & Design *UoN 2021*

- Presenting practical content and guiding students in the hands-on application of advanced concepts in research methodology, statistics, Bayesian analysis, and experimental design.

Cognitive Psychology *UoN 2020*

- Introduction and overview of fundamental cognitive psychology principles, including attention, memory, language, reasoning, and problem solving, with focus on the application of experimental methods and formal psychology report writing.

**Accreditation**
Associate *Centre for the Integration of Research, Teaching and Learning, 2024*

- Associate-level teaching accreditation via CIRTL by completing professional development events and demonstrating expertise in evidence-based teaching.

## GRANTS & FUNDING

**Enhancing Trust Calibration: An Innovative Intervention for Optimal AI-Assisted Decision Making** *2025*
Partner Investigator (Lead CI: Dr Quentin Gronau)
Pilot Research Scheme, College of Engineering, Science & Environment, University of Newcastle *AUD $13,708*

**Research conducted within funded projects**
NSF SES-2237119; NSF SES-2523723 (The Ohio State University)

## AWARDS & HONORS

**People's Choice, Best Presentation** *2024*
Virtual Mathematical Psychology Conference

**Best Poster** *2023*
UT Health Postdoctoral Research Forum

**Summer Research Internship** *2019*
University of Newcastle, Centre for Brain and Mental Health

**Best Presentation** *2018*
University of NewcastleStudent Research Conference

**Faculty of Science Commendation List** *2018*
University of Newcastle

## PROFESSIONAL SERVICE

**Conference Session Chair** *2022, 2025, 2026*
Society for Mathematical Psychology (2025, 2026)
Australian Society for Mathematical Psychology (2022)

**Conference Reviewing and Judging** *2023–2027*
Graduate abstracts and presentations judge, Hayes Graduate Research Forum (OSU)
Abstract reviewer, Southwestern Psychological Association
3-Minute Thesis judge, UT San Antonio

## UNIVERSITY SERVICE

**Founder and Chair, Postdoctoral Writing Club** *2024*
UT San Antonio

**Postdoctoral Service Roles** *2023–2024*
Postdoctoral Appreciation Week Organising Committee, UT San Antonio
Postdoctoral Mentor of the Year Judge, UT San Antonio
Postdoctoral Association Member, UT San Antonio

**Community and Governance Service** *2020–2021*
Board of Directors, Community Representative, Newcastle University Sport

## MENTORING

**Laiton G. Hedley** *University of Newcastle*
Undergraduate; now PhD candidate. First author on collaborative human–machine team-performance papers.

**Ying-Yu Chen** *UT San Antonio*
Master's student; now PhD candidate. Co-author, *Applied Configural Processing* (in prep.).

**Serena Browning** *UT San Antonio*
Undergraduate; now Master's student.

**Lauren Kahn** *UT San Antonio*
Undergraduate; now Master's student.

# PUBLICATIONS

## Peer-reviewed journal articles

Hedley, L. G., Russell, L., **Bennett, M. S.**, and Eidels, A. (2026). Benchmarking the Benchmark: A Novel Simulation Benchmark for Team Performance. *Chinese Journal of Psychology*, 68(3)

**Bennett, M. S.**, Hedley, L., Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2025). Human performance in competitive and collaborative human–machine teams. *Topics in Cognitive Science*, 17(2):324–348

Kerr, A., Carlson, S., Baldwin, Z., Innes, R., Howard, Z., **Bennett, M. S.**, Evans, T., McKenzie, T., and Dalton, C. (2025). FluTracking: Weekly online community-based surveillance of respiratory illness in Australia, 2020–2022 Report. *Communicable Diseases Intelligence*, 49

Gronau, Q. F., **Bennett, M. S.**, Brown, S. D., Hawkins, G. E., and Eidels, A. (2023). Do choice tasks and rating scales elicit the same judgments? *Journal of Choice Modelling*, 49:100437

Garrett, P. M., **Bennett, M. S.**, Hsieh, Y.-T., Howard, Z. L., Yang, C.-T., Little, D. R., and Eidels, A. (2022). Wheel of fortune: A cross-cultural examination of how expertise shapes the mental representations of familiar and unfamiliar numerals. *Computational Brain & Behavior*, 5(1):45–59

John, A. R., Singh, A. K., Do, T.-T. N., Eidels, A., Nalivaiko, E., Gavgani, A. M., Brown, S. D., **Bennett, M. S.**, Lal, S., Simpson, A. M., et al. (2022). Unraveling the physiological correlates of mental workload variations in tracking and collision prediction tasks. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:770–781

**Bennett, M. S.**, Mullard, R., Adam, M. T., Steyvers, M., Brown, S. D., and Eidels, A. (2020). Going, going, gone: competitive decision-making in Dutch auctions. *Cognitive Research: Principles and Implications*, 5(1):62

## Under Review

Alaukik†, A., **Bennett†, M. S.**, Smith, C. T., and Kvam, P. D. (2026). A road map for assessing process independence in behavioral measures of attitudes: An SFT and GRT model analysis. *Under review at Journal of Mathematical Psychology*, https://doi.org/10.31234/osf.io/jqasm_v1

**Bennett, M. S.**, Houpt, J. W., Wenger, M. J., Townsend, J. T., and Mark, L. (2026c). Scalable estimation of continuous perceptual representations from human judgements and computer vision: An application to melanoma identification. *Under review at Behavior Research Methods*, https://doi.org/10.31234/osf.io/kgeyt_v1

**Bennett, M. S.** and Kvam, P. D. (2026). Machine metacognition improves classification performance and uncertainty quantification. *Under review at Transactions on Pattern Analysis and Machine Intelligence*, https://doi.org/10.31234/osf.io/jp83f_v1

Fitch, A., Hedley, L., **Bennett, M. S.**, and Kvam, P. D. (2026). Real-time Cognitive Modeling using Simulation-Inversion Networks. *Under review at British Journal of Mathematical and Statistical Psychology*, https://doi.org/10.31234/osf.io/vbjd3_v1

### Book Chapters

Kvam, P. D., Sokratous, K., **Bennett, M. S.**, Alaukik, A., and Wang, Y. (2026). Computational modeling in judgment and decision-making. In *The Cambridge Handbook of Judgment and Decision Making*. Cambridge University Press. Invited chapter; submitted and under review

### In Preparation

**Bennett, M. S.** (2026a). Bayesian Functional Estimation of Workload Capacity in Systems Factorial Technology. *In preparation*

**Bennett, M. S.**, Houpt, J. W., and Kvam, P. D. (2026b). GRIN: Amortised Inference for General Recognition Theory. *In preparation*

**Bennett, M. S.** and Houpt, J. W. (2026). A Preprocessing and Correction Framework for Multidimensional Perceptual Judgment Data with Naturalistic Stimuli. *In preparation*

**Bennett, M. S.** (2026b). Structural Heuristics in Human and Machine Metacognition: A Representational Framework. *In preparation*

**Bennett, M. S.**, Fitch, A. K., Hintze, A., Kvam, P. D., and Sokratous, K. (2026a). Strategic behaviour in aligned and unaligned language models. *In preparation*

Houpt, J. W., **Bennett, M. S.**, Silvas, E., and Chen, Y.-Y. (2025). Applied Configural Processing. *In preparation*

### Conference Proceedings

Hedley, L. G., **Bennett, M. S.**, Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2023). The relationship between teaming behaviours and joint capacity of hybrid human-machine teams. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 45

### Reviewing Activity

Ad hoc reviewer for: *Cognitive Research: Principles and Implications, Computational Brain and Behaviour, Topics in Cognitive Science, Communications Psychology, Journal of Mathematical Psychology, Psychology and Aging*

## SOFTWARE

**GRIN** — General Recognition Inference Network. Amortised Bayesian inference for General Recognition Theory: a neural posterior estimator, trained once on simulated data, that replaces per-participant maximum-likelihood fitting of perceptual representations. Open source under an MIT licence, as a Python package (`grintools`) and an R package (`grin`) wrapping the same trained weights, and as a browser application that runs inference client-side. With J. W. Houpt and P. D. Kvam.
grin.murraysbennett.com · github.com/MurraySBennett/grin

# PRESENTATIONS

## Invited

*Human-AI perceptual judgements of malignant melanoma.* 2024, Psychology Department, The University of Newcastle, Australia.

*Human-AI collaboration.* 2024, Department of Psychological Sciences at Tarleton State University.

*Collaborative and competitive 'Team Spirit'.* 2024, Joint meeting of the Newcastle Cognition Lab and UC Irvine's MADLAB.

## Talks

**Bennett, M. S.**, Alaukik, A., Smith, C., and Kvam, P. (2025, July 25-28). Processing independence in implicit attitudes: An SFT and GRT analysis. In *Meeting of the Society for Mathematical Psychology/ICCM*

**Bennett, M. S.** and Houpt, J. W. (2024, June 17-21). A GRT examination of the ABCs for melanoma identification with comparison to a deep convolutional neural network. In *Virtual Meeting of the Society for Mathematical Psychology/ICCM*

# PROFESSIONAL SOCIETIES

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|---|---|
| Society for Mathematical Psychology | *2019-present* |
| Psychonomic Society | *2019-present* |
| Southwestern Psychological Association | *2024* |

# PUBLIC ENGAGEMENT

2021. Online article published in *The Conversation* (Australia): *Why the competitive spirit can take over in auctions and how you can stay in control.*